Abhishek Sharma

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Backend engineer with 5 years of experience in Python and Go, now focused on production LLM systems. Built and deployed a RAG assistant with model-based tool routing (embeddings, ChromaDB, FastAPI), then hardened it with live evals, prompt-injection testing, cost tracking, caching and rate limiting. At Citrix, built a Claude Code-orchestrated Jenkins failure-triage agent and an Azure OpenAI Computer Use framework that eliminated 83% of manual testing cycles. Uses AI developer tools daily and owns features from backend service to user-facing integration.

Core Competencies

- **LLM & RAG Systems**: Chunking, embeddings, vector search (ChromaDB, HNSW), retrieval debugging, model-based tool routing with rule-based fallback, structured JSON output, prompt-injection hardening, eval harnesses; Azure OpenAI (GPT-4), Computer Use API, Claude Code agentic workflows, Groq (Llama), Ollama, Hugging Face Inference API
- **LLM Reliability & Observability**: Automated evals against live endpoints, latency and token-cost tracking, TTL caching, per-IP rate limiting, input validation, graceful degradation on provider outages; structured logging (slog), Prometheus, request-ID tracing
- **Backend & APIs**: Python (FastAPI, Flask), Go (net/http, goroutines, circuit breaker, worker pool), REST API design, PostgreSQL, Redis, MongoDB, SQL
- **Delivery & Infrastructure**: Docker, GitHub Actions, Jenkins, Render deployments; cross-team exposure to multi-cloud Kubernetes on AWS (EKS), Azure (AKS) and GCP (GKE) via Terraform and Helm
- **Full-Stack**: React, Redux, TypeScript, Next.js, Node.js, Chrome Extension (MV3), JWT auth

Professional Experience

Citrix Systems — **Bengaluru, KA**
Software Engineer — *May 2023–Present*

- **AI Agents & Tooling:** Built an AI-powered Jenkins log analyzer (Claude Code + Slack API) that turns raw nightly CI failures into per-category bug tickets, with orchestrator fan-out, deduplication against existing issues and structured Slack summaries, cutting triage from hours to minutes. Also built Claude Code agents, skills, plugins and workflows for internal engineering automation.
- **LLM Automation:** Developed an Azure OpenAI + Playwright framework orchestrated by the GPT-4 Computer Use API, with self-healing browser automation that eliminated 83% of manual testing cycles across the Citrix admin portal.
- **Product Features:** Shipped admin capabilities into NetScaler Gateway for enterprise customers globally (Device Posture Service as an auth factor, EDT Lossy ICA params, Advance Policy controls) and fixed production bugs for named enterprise accounts with parallel backports across two release trains.
- **Scale & Reliability:** Resolved critical CSA script failures, cutting login failures 35% for 1M+ users; automated Jenkins CI/CD pipelines, cutting release cycle time 40%; improved React dashboard performance 40%; collaborated cross-team on multi-cloud Kubernetes provisioning with Terraform and Helm; built and verified Go features (idle-connection timeout, JSON stats output, connection cap) for an internal test tool.

Juniper Networks **Bengaluru, KA**
Software Engineer 2 *Jul 2021–May 2023*

- Built a Negative Event API in Python/Flask enabling 200+ automated test scenarios, with clean REST contracts, structured error responses and edge-case coverage for MX device deployments.
- Enhanced bandwidth monitoring and telemetry reporting for MX devices, shipping production-tested backend functionality.
- Migrated 50+ legacy scripts to Python with parallel execution, cutting test cycle time 35%.

Projects

github.com/absep98/AI-learning-Kickoff **Python · FastAPI · ChromaDB · Groq · Render**
Repo Assistant: RAG + Tool-Routing LLM App *Self-directed*

- Built and deployed a RAG assistant (697 chunks across 27 note files) with a model-based tool-routing planner that falls back to rules on invalid output, served through a FastAPI endpoint with a web chat UI. Moved embeddings from local Ollama to in-process sentence-transformers, then to a hosted API after an out-of-memory crash on a 512MB instance.
- Wrote eval harnesses for retrieval (20 questions, 5% baseline raised to 80% through retrieval debugging and eval tuning) and for the live endpoint (90% pass on 10 questions); diagnosed a chunk-splitting retrieval failure; ran six prompt-injection attempts and closed an information-disclosure gap in two prompt-hardening iterations.
- Added per-request token and cost tracking, a TTL answer cache (found and fixed a cache-poisoning bug), per-IP rate limiting, input validation, a feedback log and graceful degradation on embedding-provider outages; removed a 43-second cold start with a keep-alive ping.

github.com/absep98 **Go · PostgreSQL · Redis · Docker**
Production-Grade Go Backend API *Open Source*

- Built a REST API in Go with Redis cache-aside, a 3-state circuit breaker, exponential backoff retry and a worker pool with graceful drain; 6-layer middleware stack and 51 tests with real Postgres in GitHub Actions CI.

Chrome Web Store, VS Code Marketplace **JavaScript · LLM APIs**
TabInsights & Markdown TTS *Published*

- Shipped two published extensions: TabInsights (5.0/5 rating) and Markdown TTS (116 installs), which generates LLM narration through OpenAI, Anthropic and Groq APIs with user-supplied keys.

Education

PSG College of Technology **Coimbatore, India**
B.Tech, Information Technology *Aug 2017–Jun 2021*
GPA: 8.28/10 · PMSSS Scholar · HackerRank Certified: Java & Python